

Module 2: Basic Switch and End Device Configuration

CCNA 1: Introduction to Networks v7.0 — Identification Questionnaire

Name: _____

Date: _____

Course & Year: NETC311 / Year III

Score: _____ / 104

GENERAL INSTRUCTIONS: This questionnaire contains 104 identification-type items based thoroughly on Module 2: Basic Switch and End Device Configuration. Read each statement carefully. Identify the correct term that completes the blank (____) by selecting the best answer among choices a, b, c, or d. Write the letter of your choice on the blank provided before each number or encircle the letter. A complete Answer Key and Study Reference is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE

____ 1. The user interface of an operating system that allows users to request specific tasks through either CLI or GUI interfaces is called the ____.

- a.) kernel
- b.) shell
- c.) hardware
- d.) firmware

____ 2. The core component of an operating system that communicates directly between computer hardware and software to manage hardware resources is the ____.

- a.) shell
- b.) registry
- c.) BIOS
- d.) kernel

____ 3. The physical parts and underlying electronic circuitry of a computer or network device are collectively known as the ____.

- a.) hardware
- b.) middleware
- c.) software
- d.) firmware

____ 4. A user-friendly operating system interface that allows users to interact with a system using graphical icons, menus, and windows is a ____.

- a.) API
- b.) Kernel shell
- c.) CLI (Command Line Interface)
- d.) GUI (Graphical User Interface)

____ 5. Network operating systems on infrastructure devices like switches and routers are typically accessed and configured through a text-based ____.

- a.) web wizard only
- b.) touchscreen
- c.) CLI (Command Line Interface)
- d.) GUI

____ 6. One reason network devices rely primarily on a CLI rather than a GUI is because GUIs consume extra resources and can fail, freeze, or ____.

- a.) encrypt
- b.) re-route
- c.) authenticate
- d.) crash

____ 7. While a PC operating system enables users to use a mouse and view visual output, a CLI-based network operating system relies primarily on a technician using a ____.

- a.) keyboard
- b.) graphics tablet
- c.) stylus pen
- d.) touchpad

_____ 8. A physical management port on a Cisco network device used for out-of-band maintenance and initial device configuration is the _____.

- a.) console port
- b.) AUX port
- c.) VTY line
- d.) Ethernet port

_____ 9. The recommended method for remotely and securely connecting to the CLI of a Cisco device over a network is _____.

- a.) Telnet
- b.) TFTP
- c.) Secure Shell (SSH)
- d.) HTTP

_____ 10. An older remote terminal protocol that establishes a remote CLI connection over the network but transmits passwords and data in plaintext is _____.

- a.) HTTPS
- b.) SFTP
- c.) SSH
- d.) Telnet

_____ 11. Connecting to a network device via the console port is considered an out-of-band management method because it does not require active _____.

- a.) network services
- b.) terminal software
- c.) a serial cable
- d.) electrical power

_____ 12. Software applications such as PuTTY, Tera Term, and SecureCRT used by technicians to connect to network devices are known as _____.

- a.) terminal emulation programs
- b.) packet analyzers
- c.) debuggers
- d.) compilers

_____ 13. The basic Cisco IOS command mode that allows access to only a limited number of basic monitoring commands is called _____.

- a.) User EXEC mode
- b.) Line configuration mode
- c.) Global configuration mode
- d.) Privileged EXEC mode

_____ 14. On a Cisco IOS device named Switch, the User EXEC mode is identified by the prompt symbol ending with _____.

- a.) >
- b.) %
- c.) \$
- d.) #

_____ 15. The Cisco IOS command mode that allows access to all device commands, monitoring features, and configuration modes is _____.

- a.) Interface mode
- b.) Setup mode
- c.) User EXEC mode
- d.) Privileged EXEC mode

_____ 16. On a Cisco IOS device named Switch, the Privileged EXEC mode is identified by the prompt ending with _____.

- a.) #
- b.) :
- c.) ~
- d.) >

_____ 17. The primary configuration mode used to configure device-wide settings and access specific subconfiguration modes is called _____.

- a.) user EXEC mode
- b.) line configuration mode
- c.) interface configuration mode
- d.) global configuration mode

_____ 18. The CLI prompt displayed when entering global configuration mode on a switch named Switch is _____.

19. The subconfiguration mode used to configure console, AUX, SSH, or Telnet access lines is called _____.
a.) routing configuration mode
b.) line configuration mode
c.) interface configuration mode
d.) vlan configuration mode
20. The CLI prompt displayed when configuring a console or VTY line on a switch is _____.
a.) Switch> b.) Switch(config-if)#
c.) Switch(config)# d.) Switch(config-line)#
21. The subconfiguration mode used to configure a switch port or router physical interface is called _____.
a.) system configuration mode
b.) line configuration mode
c.) interface configuration mode
d.) device configuration mode
22. The CLI prompt displayed when configuring a switch interface such as FastEthernet or VLAN 1 is _____.
a.) Switch(config-if)# b.) Switch(config)#
c.) Switch(config-line)# d.) Switch#
23. To move from User EXEC mode to Privileged EXEC mode, the administrator enters the command _____.
a.) disable b.) enable
c.) exit d.) configure terminal
24. To move from Privileged EXEC mode down to User EXEC mode, enter the command _____.
a.) exit b.) disable
c.) logout d.) enable
25. To navigate from Privileged EXEC mode into global configuration mode, execute the command _____.
a.) configure terminal b.) enable config
c.) interface config d.) setup terminal
26. To return from global configuration mode back to Privileged EXEC mode, or from any subconfiguration mode back to global configuration mode, use the command _____.
a.) quit b.) back
c.) stop d.) exit
27. To exit from any subconfiguration mode directly back to Privileged EXEC mode in a single step, use the command _____.
a.) close b.) end
c.) return d.) exit
28. The keyboard shortcut that performs the exact same function as the 'end' command by returning directly to Privileged EXEC mode is _____.
a.) Ctrl+X b.) Ctrl+Z
c.) Ctrl+Shift+6 d.) Ctrl+C
29. In a Cisco IOS command string, a predefined parameter defined by the operating system is known as a _____.
a.) constant b.) argument

- c.) variable d.) keyword

____ **30.** In a Cisco IOS command string, a user-supplied parameter or variable (such as an IP address or password) is called an ____.

- a.) delimiter b.) operand
c.) keyword d.) argument

____ **31.** In Cisco IOS command syntax documentation, text formatted in boldface indicates commands and keywords that must be entered ____.

- a.) in uppercase only b.) as optional values
c.) literally as shown d.) as user-defined variables

____ **32.** In Cisco IOS command syntax documentation, text formatted in italics indicates an element for which the user must supply a ____.

- a.) command prompt b.) bracket
c.) literal string d.) value

____ **33.** In Cisco command syntax conventions, elements enclosed within square brackets [] represent an ____ element.

- a.) obsolete b.) optional
c.) required d.) mandatory

____ **34.** In Cisco command syntax conventions, elements enclosed within braces { } represent a ____ element.

- a.) required b.) deprecated
c.) optional d.) commented

____ **35.** In Cisco command syntax conventions, a vertical line | within braces or brackets represents a ____.

- a.)** wildcard character
b.) logical AND condition
c.) choice between mutually exclusive elements
d.) pipe to file output

____ **36.** In Cisco syntax conventions, the pattern [x {y | z}] indicates a required choice within an ____ element.

- a.) mandatory b.) optional
c.) administrative d.) encrypted

____ **37.** The Cisco IOS command used to verify layer 3 IP reachability to a destination device by sending ICMP echo requests is ____.

- a.) telnet b.) ping
c.) traceroute d.) arp

____ **38.** The Cisco IOS command used to trace the path and display intermediate router hops to a remote host is ____.

- a.) ping b.) traceroute
c.) ipconfig d.) netstat

____ **39.** Entering a question mark (?) at the CLI prompt to see which commands or arguments are available utilizes the feature called ____.

- a.) auto-complete buffer b.) command syntax check
c.) terminal logging d.) context-sensitive help

____ **40.** When an entered command is incomplete, ambiguous, or invalid, the IOS feature that provides feedback indicating the error is ____.

- a.) command syntax check b.) terminal history
c.) packet checker d.) context-sensitive help

_____ **41.** In the Cisco CLI, entering a unique starting sequence of characters and pressing the _____ key will automatically complete the command.

- a.) Tab b.) Esc
- c.) Spacebar d.) Enter

_____ **42.** In the Cisco CLI, commands can be shortened to the minimum number of characters that identify a _____ selection.

- a.) unique b.) alphabetic
- c.) numerical d.) case-sensitive

_____ **43.** The keystroke used in the CLI editing buffer to erase the character to the left of the cursor is _____.

- a.) Tab b.) Spacebar
- c.) Delete d.) Backspace

_____ **44.** The keyboard shortcut that moves the cursor one character to the left, identical to the Left Arrow key, is _____.

- a.) Ctrl+A b.) Ctrl+B
- c.) Ctrl+F d.) Ctrl+P

_____ **45.** The keyboard shortcut that moves the cursor one character to the right, identical to the Right Arrow key, is _____.

- a.) Ctrl+F b.) Ctrl+N
- c.) Ctrl+E d.) Ctrl+B

_____ **46.** The keyboard shortcut that recalls the previous command in the history buffer, identical to the Up Arrow key, is _____.

- a.) Ctrl+U b.) Ctrl+N
- c.) Ctrl+R d.) Ctrl+P

_____ **47.** When command output produces more text than can be displayed in the terminal window, the IOS pauses and displays the _____ prompt.

- a.) --Wait-- b.) --Pause--
- c.) --Continue-- d.) --More--

_____ **48.** While viewing output at the '--More--' prompt, pressing the _____ displays the next single line.

- a.) Enter key b.) Shift key
- c.) Tab key d.) Spacebar

_____ **49.** While viewing output at the '--More--' prompt, pressing the _____ displays the next full screen.

- a.) Tab key b.) Enter key
- c.) Spacebar d.) Backspace

_____ **50.** Pressing any alphanumeric key other than Enter or Spacebar while at the '--More--' prompt ends the display string and returns to _____.

- a.) User EXEC mode b.) ROMmon mode
- c.) Global configuration mode d.) Privileged EXEC mode

_____ **51.** When in any configuration mode, entering the shortcut _____ ends configuration mode and returns to privileged EXEC mode.

- a.) Ctrl-D b.) Ctrl-A
- c.) Ctrl-Shift-6 d.) Ctrl-C

_____ **52.** The all-purpose break sequence used to abort DNS lookups, traceroutes, and pings on Cisco devices is _____.

- a.) Ctrl-Alt-Delete b.) Ctrl-C
- c.) Ctrl-Shift-6 d.) Ctrl-Z

____ 53. The very first configuration command that should be applied to any Cisco device to provide a unique network identity is assigning a ____.

- a.) hostname b.) subnet mask
- c.) banner d.) IP address

____ 54. By default, an unconfigured Cisco IOS switch is assigned the factory default hostname of ____.

- a.) Default b.) Cisco
- c.) Router d.) Switch

____ 55. According to Cisco naming guidelines, a device hostname must start with a letter, contain no spaces, end with a letter or digit, and be less than ____ characters in length.

- a.) 128 b.) 64
- c.) 32 d.) 256

____ 56. To remove a configured hostname and return a switch prompt to its factory default name, enter the global command ____.

- a.) no hostname b.) clear hostname
- c.) erase hostname d.) default hostname

____ 57. Cisco password guidelines recommend that passwords used for administrative device security should be more than ____ characters in length.

- a.) six b.) eight
- c.) sixteen d.) four

____ 58. Simple passwords such as 'cisco' or 'class' used in lab environments are considered weak and easily guessable, and should be avoided in ____ environments.

- a.) production b.) classroom
- c.) simulation d.) virtual

____ 59. To configure user EXEC password security for the physical management console, the administrator enters the global command ____.

- a.) line terminal 0 b.) line aux 0
- c.) line vty 0 d.) line console 0

____ 60. In line configuration mode, after entering the 'password <string>' command, user EXEC access must be enabled using the command ____.

- a.) enable b.) login
- c.) activate d.) auth

____ 61. To secure Privileged EXEC mode access with an encrypted, hashed password, the administrator uses the global command ____.

- a.) encrypt enable b.) enable secret
- c.) enable password d.) service password

____ 62. Virtual lines that allow remote administrative access using Telnet or SSH to a Cisco device are called ____ lines.

- a.) Ethernet b.) Console
- c.) VTY d.) AUX

____ 63. Many Cisco switches support up to 16 virtual terminal lines, which are configured concurrently using the command ____.

- a.) line vty 0 15 b.) line remote 0 15
- c.) line console 0 15 d.) line vty 1 16

____ 64. By default, configuration files such as startup-config and running-config display configured passwords in ____.

- a.) plaintext
- b.) MD5 hashes
- c.) hexadecimal codes
- d.) SHA-256 ciphertext

_____ 65. To encrypt all plaintext passwords stored in the configuration files, execute the global configuration command _____.

- a.) encrypt-passwords all
- b.) crypto password-encrypt
- c.) service password-encryption
- d.) secure passwords enable

_____ 66. To verify that all configured passwords on the device are now encrypted, the administrator executes the Privileged EXEC command _____.

- a.) show security-keys
- b.) show passwords
- c.) show encryption
- d.) show running-config

_____ 67. A legal notification displayed to warn unauthorized personnel against attempting to access a device is configured using the global command _____.

- a.) banner motd
- b.) login prompt
- c.) legal notice
- d.) syslog warning

_____ 68. In the command 'banner motd # Unauthorized Access Prohibited #', the '#' character placed at the beginning and end of the message is called the _____ character.

- a.) delimiting
- b.) control
- c.) wildcard
- d.) escape

_____ 69. The saved configuration file stored in NVRAM that is loaded and executed when a Cisco device reboots is called _____.

- a.) startup-config
- b.) running-config
- c.) active-config
- d.) boot-config

_____ 70. The active configuration file stored in Random Access Memory (RAM) that immediately reflects changes made on the CLI is the _____.

- a.) flash-config
- b.) startup-config
- c.) nvram-config
- d.) running-config

_____ 71. Because system RAM is _____ memory, modifying the running configuration affects operation immediately, but changes are lost if power is lost.

- a.) permanent
- b.) static
- c.) non-volatile
- d.) volatile

_____ 72. The persistent storage location where the startup configuration file is preserved across reboots and power outages is _____.

- a.) NVRAM
- b.) TFTP server
- c.) ROM cache
- d.) RAM

_____ 73. To save all active modifications from volatile RAM to non-volatile NVRAM, enter the Privileged EXEC command _____.

- a.) copy running-config startup-config
- b.) save running-config startup-config
- c.) commit startup-config
- d.) write ram nvram

_____ 74. If undesirable changes have been made to the running configuration and have not yet been saved, the administrator can restore the previous saved configuration using the command _____.

- a.) reload
- b.) reboot system
- c.) reset factory
- d.) rollback

____ 75. If unwanted configuration changes have already been saved to the startup configuration file, the administrator can clear NVRAM using the command ____.

- a.) delete running-config b.) reset startup-config
- c.) clear nvram-all d.) erase startup-config

____ 76. After clearing the startup configuration file from NVRAM with 'erase startup-config', the administrator must enter the ____ command to clear the running-config from RAM.

- a.) shutdown b.) exit
- c.) reload d.) erase running-config

____ 77. When archiving a Cisco device configuration to a text file using terminal software such as PuTTY or Tera Term, the first step is to enable session ____.

- a.) scripting b.) mirroring
- c.) sniffing d.) logging

____ 78. While session logging is active in terminal software, the administrator captures the active configuration by issuing the Privileged EXEC command 'show running-config' or ____.

- a.) show flash b.) show interfaces
- c.) show startup-config d.) show memory

____ 79. After capturing configuration output in terminal software, logging is turned off by choosing the session logging option ____.

- a.) Disable b.) Stop
- c.) None d.) Close

____ 80. The primary means of enabling devices to locate one another and establish end-to-end communication on the Internet is using ____.

- a.) MAC addresses only b.) port numbers only
- c.) hostnames only d.) IP addresses

____ 81. The structure of an IPv4 address is formatted as four decimal numbers between 0 and 255 separated by dots, known as ____.

- a.) hexadecimal notation b.) dotted decimal notation
- c.) colon-hex format d.) binary octet notation

____ 82. In an IPv4 address, each of the four decimal numbers must have a value between 0 and ____.

- a.) 512 b.) 255
- c.) 127 d.) 256

____ 83. An IPv4 subnet mask is a 32-bit value that differentiates the network portion of the address from the ____ portion.

- a.) interface b.) host
- c.) broadcast d.) domain

____ 84. The IP address of the local router interface that an end host uses to reach remote networks and the internet is the ____.

- a.) subnet mask b.) default gateway
- c.) broadcast address d.) DNS server

____ 85. An IPv6 address has a total length of ____ bits.

- a.) 256 b.) 128
- c.) 32 d.) 64

____ 86. An IPv6 address is written as a sequence of hexadecimal values separated into groups of four digits by the symbol ____.

- a.) colon (:) b.) hyphen (-)

- c.) slash (/) d.) period (.)

____ **87.** In an IPv6 address, each individual hexadecimal digit represents ____ binary bits.

- a.) 8 b.) 16
c.) 4 d.) 32

____ **88.** A complete IPv6 address contains a total of ____ hexadecimal digits.

- a.) 16 b.) 32
c.) 128 d.) 64

____ **89.** Unlike some command parameters, IPv6 addresses are not case-sensitive and can be written in either lowercase or ____.

- a.) octal b.) uppercase
c.) binary d.) italics

____ **90.** Differences between network media types include transmission speed, cost, environment, and the distance the media can carry a ____.

- a.) signal b.) VLAN
c.) gateway d.) protocol

____ **91.** End devices on a network require an IP address in order to communicate with other devices, which can be configured manually or automatically using ____.

- a.) ARP b.) DHCP
c.) ICMP d.) DNS

____ **92.** To manually configure an IPv4 address on a Windows PC, a user opens Control Panel > Network Sharing Center > Change adapter settings, selects the adapter, and opens ____ Properties.

- a.) Local Area Connection b.) Administrative Tools
c.) Device Manager d.) Windows Defender

____ **93.** In Windows PC networking, manual IPv4 settings are entered within the Properties dialog of Internet Protocol Version 4 (____).

- a.) UDP/IPv4 b.) TCP/IPv4
c.) NETBIOS/IPv4 d.) ICMP/IPv4

____ **94.** The network protocol that automatically assigns an IPv4 address, subnet mask, default gateway, and DNS servers to end devices is ____.

- a.) DNS b.) NTP
c.) DHCP d.) SMTP

____ **95.** By default, end devices such as PCs are configured to obtain an IP address automatically using ____.

- a.) DHCP b.) Static IP
c.) APIPA only d.) Manual assignment

____ **96.** In addition to DHCPv6, an IPv6 end device can automatically acquire dynamic address allocation using ____.

- a.) NAT**
b.) Proxy DNS
c.) SLAAC (Stateless Address Autoconfiguration)
d.) ARP

____ **97.** To enable remote management access (SSH or Telnet) to a Layer 2 switch across a network, an IP address must be configured on an ____.

- a.) auxiliary line**
b.) physical loopback cable
c.) console port

d.) SVI (Switch Virtual Interface)

____ **98.** The default Switch Virtual Interface present on Cisco switches used for remote IP management is ____.

- a.) interface loopback 0 b.) interface fastethernet 0/1
- c.) interface vlan 10 d.) interface vlan 1

____ **99.** To configure an IP address on an SVI, the administrator enters the command 'ip address' followed by the ip-address and the ____.

- a.) mac-address b.) default-gateway
- c.) subnet-mask d.) dns-server

____ **100.** After configuring the IP address and subnet mask on an SVI, the interface must be administratively enabled using the command ____.

- a.) no shutdown b.) start
- c.) up d.) enable interface

____ **101.** On a Windows PC command prompt, the command used to verify the current IP address, subnet mask, and default gateway configuration is ____.

- a.) ifconfig b.) show ip
- c.) netstat d.) ipconfig

____ **102.** On a Cisco switch CLI, the Privileged EXEC command used to display a concise summary of the status and IP configuration of all interfaces is ____.

- a.) show ip route**
- b.) show interfaces status**
- c.) show running-config interface**
- d.) show ip interface brief**

____ **103.** To navigate from global configuration mode into interface configuration mode for the management VLAN, enter the command ____.

- a.) interface vlan 0 b.) switchport vlan 1
- c.) interface vlan 1 d.) line vlan 1

____ **104.** The command mode accessed immediately when a user establishes a console connection or boots a Cisco switch without entering passwords is ____.

- a.) Line configuration mode b.) Privileged EXEC mode
- c.) Global configuration mode d.) User EXEC mode

PART II: ANSWER KEY & STUDY REFERENCE — MODULE 2: BASIC SWITCH AND END DEVICE CONFIGURATION

Official Solutions, Correct Terms, and Curriculum Concept References

Item #	Key	Correct Identification Term	Curriculum Topic & Reference Explanation
1	b.)	shell	[2.1 Operating Systems] The shell is the user interface that allows users to request specific tasks from the computer via CLI or GUI.
2	d.)	kernel	[2.1 Operating Systems] The kernel communicates between the hardware and software of a computer and manages how resources are used.
3	a.)	hardware	[2.1 Operating Systems] Hardware represents the physical parts of a computer including underlying electronics.
4	d.)	GUI (Graphical User Interface)	[2.1 GUI] A GUI allows the user to interact with the system using graphical icons, menus, and windows.
5	c.)	CLI (Command Line Interface)	[2.1 GUI] Network devices are typically accessed through a CLI because GUIs can crash, fail, or consume unnecessary resources.
6	d.)	crash	[2.1 GUI] GUIs can fail, crash, or simply not operate as specified; network devices are typically accessed through CLI.
7	a.)	keyboard	[2.1 Purpose of an OS] A CLI-based network OS enables a network technician to use a keyboard to run CLI-based network programs and enter text commands.
8	a.)	console port	[2.1 Access Methods] The console port is a physical management port used to access a device for maintenance, such as initial configuration.
9	c.)	Secure Shell (SSH)	[2.1 Access Methods] SSH establishes a secure remote CLI connection through a virtual interface over a network with encryption.

10	d.)	Telnet	[2.1 Access Methods] Telnet establishes an insecure remote CLI connection where user authentication, passwords, and commands are sent in plaintext.
11	a.)	network services	[2.1 Access Methods] Console access does not require any network connectivity or IP configuration on the device.
12	a.)	terminal emulation programs	[2.1 Terminal Emulation Programs] Terminal emulation programs are used to connect to a network device by either a console port or SSH/Telnet.
13	a.)	User EXEC mode	[2.2 Primary Command Modes] User EXEC mode allows access to only a limited number of basic monitoring commands.
14	a.)	>	[2.2 Primary Command Modes] User EXEC mode is identified by the CLI prompt that ends with the > symbol (e.g., Switch>).
15	d.)	Privileged EXEC mode	[2.2 Primary Command Modes] Privileged EXEC mode allows access to all commands and features and is identified by the # prompt.
16	a.)	#	[2.2 Primary Command Modes] Privileged EXEC mode is identified by the CLI prompt ending with the # symbol (e.g., Switch#).
17	d.)	global configuration mode	[2.2 Configuration Modes] Global configuration mode is used to access configuration options on the device.
18	d.)	Switch(config)#	[2.2 Configuration Modes] Global configuration mode is designated by the prompt Switch(config)#.
19	b.)	line configuration mode	[2.2 Subconfiguration Modes] Line configuration mode is used to configure console, SSH, Telnet or AUX access.
20	d.)	Switch(config-line)#	[2.2 Subconfiguration Modes] Line configuration mode displays the prompt Switch(config-line)#.

21	c.)	interface configuration mode	[2.2 Subconfiguration Modes] Interface configuration mode is used to configure a switch port or router interface.
22	a.)	Switch(config-if)#	[2.2 Subconfiguration Modes] Interface configuration mode displays the prompt Switch(config-if)#.
23	b.)	enable	[2.2 Navigation Between Modes] The enable command transitions the CLI from User EXEC mode to Privileged EXEC mode.
24	b.)	disable	[2.2 Navigation Between Modes] The disable command returns the CLI from Privileged EXEC mode to User EXEC mode.
25	a.)	configure terminal	[2.2 Navigation Between Modes] The configure terminal command moves the device from Privileged EXEC into global configuration mode.
26	d.)	exit	[2.2 Navigation Between Modes] The exit command returns to the immediately preceding command mode.
27	b.)	end	[2.2 Navigation Between Modes] The end command returns directly to Privileged EXEC mode from any configuration mode.
28	b.)	Ctrl+Z	[2.2 Navigation Between Modes] The key combination Ctrl+Z ends any configuration mode and returns to privileged EXEC mode.
29	d.)	keyword	[2.3 Basic IOS Command Structure] A keyword is a specific parameter defined in the operating system (e.g., 'ip protocols').
30	d.)	argument	[2.3 Basic IOS Command Structure] An argument is not predefined; it is a value or variable defined by the user (e.g., '192.168.10.5').
31	c.)	literally as shown	[2.3 IOS Command Syntax Check] Boldface text indicates commands and keywords that you enter literally as shown.
32	d.)	value	[2.3 IOS Command Syntax Check] Italic text indicates

			arguments for which you supply values.
33	b.)	optional	[2.3 IOS Command Syntax Check] Square brackets [x] indicate an optional element (keyword or argument).
34	a.)	required	[2.3 IOS Command Syntax Check] Braces {x} indicate a required element (keyword or argument).
35	c.)	choice between mutually exclusive elements	[2.3 IOS Command Syntax Check] A vertical line indicates a required or optional choice between mutually exclusive elements.
36	b.)	optional	[2.3 IOS Command Syntax Check] Braces and vertical lines within square brackets indicate a required choice within an optional element.
37	b.)	ping	[2.3 IOS Command Syntax Check] The ping command sends ICMP echo requests to test connectivity to an IP address.
38	b.)	tracert	[2.3 IOS Command Syntax Check] The tracert command traces the route and lists each intermediate router hop to a destination.
39	d.)	context-sensitive help	[2.3 IOS Help Features] Context-sensitive help allows users to see available commands, keywords, and arguments by typing '?'. [2.3 IOS Help Features] Command syntax check verifies that a valid command was entered and gives feedback if invalid.
40	a.)	command syntax check	
41	a.)	Tab	[2.3 Hot Keys and Shortcuts] The Tab key completes a partial command name entry.
42	a.)	unique	[2.3 Hot Keys and Shortcuts] Commands and keywords can be shortened to the minimum number of characters that identify a unique selection.
43	d.)	Backspace	[2.3 Hot Keys and Shortcuts] Backspace erases the character to the left of the cursor.

44	b.)	Ctrl+B	[2.3 Hot Keys and Shortcuts] Left Arrow or Ctrl+B moves the cursor one character to the left.
45	a.)	Ctrl+F	[2.3 Hot Keys and Shortcuts] Right Arrow or Ctrl+F moves the cursor one character to the right.
46	d.)	Ctrl+P	[2.3 Hot Keys and Shortcuts] Up Arrow or Ctrl+P recalls commands in the history buffer, beginning with the most recent.
47	d.)	--More--	[2.3 Hot Keys and Shortcuts] When output exceeds the terminal window, the IOS displays a '--More--' prompt.
48	a.)	Enter key	[2.3 Hot Keys and Shortcuts] At the '--More--' prompt, the Enter key displays the next line.
49	c.)	Spacebar	[2.3 Hot Keys and Shortcuts] At the '--More--' prompt, the Spacebar displays the next full screen.
50	d.)	Privileged EXEC mode	[2.3 Hot Keys and Shortcuts] Pressing any other key ends the display string, returning to privileged EXEC mode.
51	d.)	Ctrl-C	[2.3 Hot Keys and Shortcuts] Ctrl-C ends configuration mode and returns to privileged EXEC mode.
52	c.)	Ctrl-Shift-6	[2.3 Hot Keys and Shortcuts] Ctrl-Shift-6 is the all-purpose break sequence used to abort DNS lookups, traceroutes, pings, etc.
53	a.)	hostname	[2.4 Device Names] The first configuration command on any device should be to give it a unique hostname.
54	d.)	Switch	[2.4 Device Names] By default, all Cisco IOS switches are assigned the factory default name 'Switch'.
55	b.)	64	[2.4 Device Names] Hostnames must start with a letter, have no spaces, end with letter/digit, and be less than 64 characters.
56	a.)	no hostname	[2.4 Device Names] To return the switch to the default

			prompt, use the 'no hostname' global config command.
57	b.)	eight	[2.4 Password Guidelines] Cisco recommends passwords that are more than eight characters in length.
58	a.)	production	[2.4 Password Guidelines] Simple passwords like 'cisco' or 'class' are weak and should be avoided in production environments.
59	d.)	line console 0	[2.4 Configure Passwords] To secure user EXEC mode access via console, first enter line console 0 configuration mode.
60	b.)	login	[2.4 Configure Passwords] The 'login' command enables password checking upon connecting to the line.
61	b.)	enable secret	[2.4 Configure Passwords] The 'enable secret password' command provides encrypted password protection for privileged EXEC mode.
62	c.)	VTY	[2.4 Configure Passwords] VTY (Virtual Terminal) lines enable remote access using Telnet or SSH to the device.
63	a.)	line vty 0 15	[2.4 Configure Passwords] Switches support 16 VTY lines numbered 0 to 15, entered using 'line vty 0 15'.
64	a.)	plaintext	[2.4 Encrypt Passwords] The startup-config and running-config files display most passwords in plaintext by default.
65	c.)	service password-encryption	[2.4 Encrypt Passwords] The 'service password-encryption' command prevents passwords from appearing as plaintext in config files.
66	d.)	show running-config	[2.4 Encrypt Passwords] Use 'show running-config' to verify that passwords in the active configuration are encrypted.
67	a.)	banner motd	[2.4 Banner Messages] A banner message of the day is configured using 'banner motd # message #' to warn unauthorized users.

68	a.)	delimiting	[2.4 Banner Messages] The character entered before and after the banner message is called the delimiting character.
69	a.)	startup-config	[2.5 Configuration Files] The startup-config is stored in NVRAM and contains commands executed upon reboot.
70	d.)	running-config	[2.5 Configuration Files] The running-config is stored in RAM and reflects the current, active device configuration.
71	d.)	volatile	[2.5 Configuration Files] RAM is volatile memory; it loses all of its contents when the device is powered off or restarted.
72	a.)	NVRAM	[2.5 Configuration Files] NVRAM (Non-Volatile RAM) does not lose its contents when the device is powered off.
73	a.)	copy running-config startup-config	[2.5 Configuration Files] The 'copy running-config startup-config' command saves the running configuration to NVRAM.
74	a.)	reload	[2.5 Alter the Running Configurations] Issuing 'reload' in Privileged EXEC mode reboots the device and reloads startup-config from NVRAM.
75	d.)	erase startup-config	[2.5 Alter the Running Configurations] The 'erase startup-config' command clears the saved configuration file from NVRAM.
76	c.)	reload	[2.5 Alter the Running Configurations] After erasing startup-config, reloading the device clears the running-config from RAM.
77	d.)	logging	[2.5 Capture Configuration to a Text File] Step 2 in capturing config is to enable session logging in the terminal emulation software.
78	c.)	show startup-config	[2.5 Capture Configuration to a Text File] Executing 'show running-config' or 'show startup-config' prints the

			configuration into the log file.
79	c.)	None	[2.5 Capture Configuration to a Text File] Session logging is disabled in PuTTY by choosing the 'None' session logging option.
80	d.)	IP addresses	[2.6 IP Addresses] IP addresses are the primary means enabling devices to locate one another and establish communication.
81	b.)	dotted decimal notation	[2.6 IP Addresses] The structure of an IPv4 address is called dotted decimal notation.
82	b.)	255	[2.6 IP Addresses] Each decimal octet in an IPv4 address ranges from 0 to 255.
83	b.)	host	[2.6 IP Addresses] The subnet mask differentiates the network portion of the address from the host portion.
84	b.)	default gateway	[2.6 IP Addresses] The default gateway is the IP address of the router used by the host to access remote networks.
85	b.)	128	[2.6 IP Addresses] IPv6 addresses are 128 bits in length.
86	a.)	colon (:)	[2.6 IP Addresses] Groups of four hexadecimal digits in IPv6 are separated by a colon ':'.
87	c.)	4	[2.6 IP Addresses] Every four bits is represented by a single hexadecimal digit.
88	b.)	32	[2.6 IP Addresses] A 128-bit IPv6 address consists of 32 hexadecimal values (128 / 4 = 32).
89	b.)	uppercase	[2.6 IP Addresses] IPv6 addresses are not case-sensitive and can be written in lowercase or uppercase.
90	a.)	signal	[2.6 Interfaces and Ports] Differences include distance the media can successfully carry a signal, speed, environment, and cost.
91	b.)	DHCP	[2.7 Manual IP Address Configuration] IPv4 address information can be entered manually or automatically

			using DHCP.
92	a.)	Local Area Connection	[2.7 Manual IP Address Configuration] Right-clicking the adapter opens the Local Area Connection Properties window.
93	b.)	TCP/IPv4	[2.7 Manual IP Address Configuration] Settings are entered in the Internet Protocol Version 4 (TCP/IPv4) Properties window.
94	c.)	DHCP	[2.7 Automatic IP Address Configuration] DHCP (Dynamic Host Configuration Protocol) enables automatic IPv4 address configuration.
95	a.)	DHCP	[2.7 Automatic IP Address Configuration] End devices are typically by default using DHCP for automatic IPv4 address configuration.
96	c.)	SLAAC (Stateless Address Autoconfiguration)	[2.7 Automatic IP Address Configuration] IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.
97	d.)	SVI (Switch Virtual Interface)	[2.7 Switch Virtual Interface Configuration] To access the switch remotely, an IP address and subnet mask must be configured on the SVI.
98	d.)	interface vlan 1	[2.7 Switch Virtual Interface Configuration] On a Cisco switch, the default virtual management interface is interface vlan 1.
99	c.)	subnet-mask	[2.7 Switch Virtual Interface Configuration] The syntax is: 'ip address ip-address subnet-mask'.
100	a.)	no shutdown	[2.7 Switch Virtual Interface Configuration] The virtual interface is enabled using the 'no shutdown' interface command.
101	d.)	ipconfig	[2.9 Module Practice and Quiz / Commands] The 'ipconfig' command displays the IP configuration on a Windows PC.
102	d.)	show ip interface brief	[2.9 Module Practice and Quiz / Commands] 'show ip

			interface brief' (or 'show ip int brief') lists interfaces, IP addresses, and line status.
103	c.)	interface vlan 1	[2.7 Switch Virtual Interface Configuration] The command 'interface vlan 1' enters SVI configuration mode.
104	d.)	User EXEC mode	[2.2 Primary Command Modes] A user lands in User EXEC mode (prompt '>') upon initial connection.